Tutorial - 9

In this tutorial, you will design and debug a couple of numerical methods written in OpenACC.

1. Consider N cities each containing maximum N localities. These can be represented using an $N \times N$ matrix. Each locality ll in the city cc has a certain number of infected people given by matrix$[cc][ll]$. The given program tries to compute the percentage of infected people in each city. Find out and fix all the errors in the given code. Introduce/correct appropriate parallel directives, clauses and data directives.

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

#define N   100
#define PROBZERO  33

int main() {
   int a[N][N];    // cities X localities
   int x[N],       // sum of infected people for all the localities of a city
   y[N];           // number of localities in a city

   srand(time(NULL));

   #pragma acc parallel
   for (int ii = 0; ii < N; ++ii) {
      x[ii] = 0;
      for (int jj = 0; jj < N; ++jj) {
         a[ii][jj] = rand() % N; //ii * N + jj;
         if (rand() % 100 < PROBZERO) a[ii][jj] = 0;  // add some zeros to make life
    interesting.
      }
   }

   #pragma acc parallel collapse(2) reduction
   for (int ii = 0; ii < N; ++ii) {
      for (int jj = 0; jj < N; ++jj) {
         x[ii] += a[ii][jj];
         if (a[ii][jj] > 0) y[ii]++;
      }
   }

   for (int ii = 0; ii < N; ++ii)
   printf("%.0f ", x[ii] * 100.0 / (y[ii] * N));
   printf("\n");

   return 0;
}
```

2. Implement an OpenACC version of matrix multiplication and demonstrate that it is producing the same result as that of the serial program.

3. Using the fourth-order Padé scheme and the third-order boundary schemes discussed in the class differntiate the following fucntion using 15 equally spaced points and plot the result obtained and compare the same with the exact derivative. The TDMA algorithm discussed in the class needs to be implmented only in serial. Compare these results with a CDS2 in the interior and FD1 and BD1 schemes at the boundaries.

$$f(x) = \sin(5x), \quad 0 \leq x \leq 3 \tag{1}$$